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Mitigating urban light pollution:
A review of municipal regulations and implications for planners

By

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Abstract

Propelled by more energy-efficient technologies such as light-emitting diodes, the growth of outdoor light at night is raising concerns about light pollution (LP). Given mounting evidence LP's risks to human wellbeing, nature and wildlife, and views of the dark sky, this paper examines the extent to which LP concerns are addressed in land use plans, policies, and regulations in nine U.S. cities, including Flagstaff, Arizona, a city known for its model lighting code. Through a review of zoning and other ordinances, area and sectoral plans and policies, media reports, and interviews with planners, this study documents that LP has yet to be taken seriously in municipal codes of most study cities. Drawing from the Flagstaff model, it offers recommendations for evidence-based, objective standards for three key dimensions of LP-mitigation: limiting total illumination or lumens per acre, shielding of lamps, and specifying luminaire characteristics.

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Biography
Kameshwari (Kami) Pothukuchi is Distinguished Service Professor of Urban Studies and Planning at Wayne State University. Her research interests include planning for community food systems, equitable community development, and nature-friendly cities.

Mitigating urban light pollution:
A review of municipal regulations and implications for planning

Artificial outdoor light at night has greatly contributed to the economic, social, and cultural life of cities over the last three hundred years and continues to expand worldwide at the rate of about 2% per year (Kyba *et al*., 2017). Energy-efficient technologies such as light-emitting diodes (or LEDs) are contributing to higher levels of nighttime illumination, a welcome development in previously underserved areas, but one that also raises concerns about light pollution (LP), or the cumulative adverse impact of excess, obtrusive light. Indeed, evidence is mounting on the risks posed by LP to human health and safety, health of nature and wildlife, and climate change (Pothukuchi, 2021). Over the last few decades, "dark-sky" advocates have called for specific policies and practices to mitigate LP.

To what extent, then, are LP concerns addressed by land use and related plans, policies, and regulations (collectively termed regulations from here on) in U.S. cities, if they are, and how, are questions taken up in this exploratory study. What rationales, purposes, and adverse impacts guide regulations and how are conflicts addressed? How do planners and elected officials view light pollution and what community-level factors support or hinder related decision-making and implementation? Finally, what recommendations emerge for planning practice and scholarship? I answer these questions through a mixed-method study that includes a review of municipal plans, policies and codes such as those related to lighting, recreation, sustainability, energy, and climate, in eight U.S. cities; a "best-practice" case-study; interviews with local officials and experts; and a scan of local news outlets for LP related awareness, information, and opinion.

In a first study of its kind, I argue that planners need to pay attention to this topic, have important roles to play and are well-situated for these roles, and can support a multitude of

actions. However, for a variety of reasons, planning lags the knowhow and technology currently available to mitigate light pollution. Although codification requires a degree of familiarity with lighting's multiple impacts and desired parameters, it is not only feasible but also shown to produce measurable impact in Flagstaff, the model case documented in this paper.

The paper is organized into four major sections. The first briefly traces urban lighting's history to showcase its growing importance as a municipal concern, summarizes key impacts of concern related to LP, and discusses rationales for planning. The following section describes the research methods used while a third describes and discusses findings. The paper concludes with recommendations for planning practice and scholarship.

Urban lighting, light pollution, and planning rationales

By 1700, public lighting was established in Amsterdam, Paris, London, and Copenhagen and elsewhere in Europe, serving mostly elite interests in trade, travel, and socialization (Koslofsky, 2002, p. 744). The first street lighting consisted of oil lamps in glass-paned lanterns attached to a post or bracket. London's early regulations required residents to hang lanterns outside their houses, and specified size, hours, days of the month (depending on moonlight), and months of the year (straddling winter), penalties for non-compliance, and exemptions (De Beer, 1941). Oil was replaced by gas in the early 19th century and, towards the end of the century, gas itself was replaced by electricity (Bouman, 1987). In New York City, Edison developed the first power company to provide current for lights, which were installed in 1882. By 1887, Los Angeles had 36 150-ft-high arc-light masts (SurveyLA 2017). Electric incandescent light soon took off nationwide; demand for light fed the production of power and vice versa (Ploetz, 2002).

Municipal governments increasingly adopted responsibility and standards for lighting the entire city and developed varied forms of financing (De Beer, 1941; Bouman, 1987).

As night time work and leisure expanded, crime-protection grew as an imperative in particular showplace districts, and business competition and civic boosterism demanded more, and more spectacular, lighting (Bouman, 1987, pp. 12-14). For example, like other U.S. cities, Minneapolis had privately provided gas lighting since 1870. After 1892, when the city took over electric lighting, the central business district, major avenues, and older neighborhoods had more lamps. Businesses along Hennepin and Nicollet Streets paid for half of the cost of the ornamental lighting installed privately to promote the businesses. A 1913 article lauded the replacement of the "ungainly post of iron with its single light" on Nicollet with the "ornamental lamp post of splendid design with five lights" (Bouman, 1987, pp. 26-28). By the turn of the 19th century, lighting was expanded to serve property values, civic beautification, recreation and entertainment, and with the advent of motorized transportation around World War I, even more road safety. It is worth noting that public lighting's development disproportionately benefited elite propertied and business interests (Bouman, 1987).

This brief history points to the growth of urban outdoor lighting in terms of illumination, geography, rationales and interests, technological innovation, and financing. It also highlights the central role of local governments in this growth. At the same time, local efforts to regulate light's more intrusive qualities also emerged, albeit more sporadically (Ploetz, 2002). Traditionally, local laws mandate that emitted light does not extend beyond the property line or create significant glare, codifying such impacts as trespass or nuisance (Ploetz, 2002, p. 1026). This is keeping with zoning's broader purpose of allowing property owners to derive benefit from the property without creating a nuisance for neighbors, and generally securing public health, safety,

and welfare. More recently, with greater awareness of light pollution and related sky glow (discussed below), a few municipalities and states have enacted regulations specifically to protect astronomical activity, endangered wildlife, and parklands, and to conserve energy (Ploetz, 2002; City of Flagstaff, AZ, n.d.; City of San Diego, CA, n.d.). Flagstaff's "best-practice" measures are discussed in a subsequent section.

The earth is getting brighter at night. Kyba *et al*. (2017) document a 2.2% average annual increase in artificially lit outdoor area with a total growth of 1.8% in upward radiance per year, albeit with large differences across regions. Light pollution generally comprises the unwanted, unintended, and obtrusive aspects of artificial lighting. The International Dark-Sky Association (IDA) defines it as 'any adverse effect of artificial light' (Stone, 2017, p. 285), while Youyuenyong (2015) calls it "every form of artificial light in the wrong place at the wrong time." About 83% of the world's population and more than 99% of the U.S. and European populations live under light-polluted skies, defined by Falchi *et al*. *(*2016, p. 4)[1] as places where artificial sky brightness at the zenith is greater than 14 mcd/m.[2] Stated more plainly, we can only see about 50 stars without any aids in many metropolitan areas compared with up to 3,000 stars in dark, rural areas (Mizon, 2012).

A community's level of local economic development—employment in manufacturing and retail and wholesale trade, for example—along with its population, income, and density are correlated with LP, as are its local roadways (Olsen *et al*., 2014). Many cities are adding new forms of outdoor illumination, including animated advertising signs and decorative lighting, to encourage tourism and nightlife (Ngarambe & Kim, 2018; Zielinska-Dabkowska & Xavia, 2019). A recent review concludes that LP has received scant attention by planners and highlights

a range of adverse impacts (Pothukuchi, 2021). A few are summarized below. The International Dark-Sky Association—or IDA—(n.d.-a) offers a wide-ranging compilation of related research.

Select adverse impacts of light pollution

The American Medical Association (2016) has designated LP a public health hazard. Cupertino *et al* (2022) summarize research on the adverse human health effects of artificial indoor and outdoor light. They cite disruptions to sleep and mood, circadian rhythms and melatonin production; activation of stress hormones and related impacts such as increased risk of certain cancers and cardio-vascular disease. Additional impacts include loss and degradation of night-vision (Pottharst and Könecke, 2013). Spectrum, illumination level measured at the eye, duration of exposure, timing of exposure relative to one's circadian cycle, and animation, all are believed to contribute to light's effects on health, sleep, mood, vigor, and overall wellbeing (see Schulte-Romer *et al*., 2019, for example). Higher levels of blue light in LED (Light-Emitting Diode) lamps exacerbate these disruptions due to LED's widespread adoption. Municipal lighting retrofits with LEDs in cities such as Seattle, WA, Davis, CA, and New York, NY, with higher blue content relative to the previous High Pressure Sodium (HPS) standard, for example, resulted in widespread resident complaints (AMA, 2016; Chaban, 2015; Scigliano, 2013). More recently, in accordance with AMA guidance, many cities are using 3000K or lower color-correlated temperature (CCT)[2] fixtures, some with dimming measures (DOE 2017, p. 15). Increasingly, billboards alongside highways and large signs with dynamic lighting are raising concerns related to driver distraction and glare (Ngarambe and Kim 2018; Zielinska-Dabkowska and Xavia 2019).

Artificial light at night has wide-ranging impacts on fish, birds, amphibians, reptiles and mammals, and indeed, entire ecological communities, with especially detrimental consequences

for threatened and endangered species (Commonwealth of Australia, 2020; Russart & Nelson 2018; Gaston *et al*., 2018; and many others). These biodiversity threats apply at multiple scales, from the neighborhood to national parks and multi-national regions. LP can change behavior and/or physiology of organisms, thereby reducing survivability or reproductive output. It can also have the indirect effect of changing the availability of habitat or food resources, and it can attract predators and invasive pests. In plants, artificial light affects growth, timing of flowering, and resource allocation, with cascading effects for pollinators and herbivores (Bennie *et al*., 2017; Yu *et al*., 2019). Indeed, some scientists believe LP to be catastrophic to ecosystems to a degree similar to climate change itself (Berthaume, 2007, p. 57). LEDs pose special threats to wildlife because many species perceive light beyond the blue range and into the ultraviolet range (Cabrera-Cruz *et al*., 2018; Kinzey *et al*., 2019). To illustrate, a study in Chile documented that approximately 20,000 Markham's storm petrel fledglings died hitting lamp posts, disoriented by the 6600K LED lamps that replaced older ones (Schulte-Römer *et al*., 2018, p. 51).

Finally, excess light wastes energy and money and exacerbates climate change (Stone, 2017; Li *et al*., 2019**)**. IDA (n.d.-b) estimates that at least 30% of all outdoor lighting in the U.S. is wasted, costing $3.3 billion and releasing 21 million tons of carbon dioxide a year. Thus, simply eliminating wasteful light can contribute to sustainability without affecting intended service levels. LEDs indeed offer opportunities for LP mitigation because they can be adapted for better optical control—that is, applying light more efficiently to the target area, dimming during low-usage times, and manipulating spectral characteristics so as to minimize incidental negative impacts. However, decisions to adopt LEDs, including for streetlight retrofits in cities, are almost entirely driven by LEDs' energy-efficiency and longevity, and thus significant cost-savings (Keatts, 2015). Because of high initial costs, some authorities lock into place long term

contracts for LED systems with energy companies or streetlight manufacturers, without a full consideration of their adverse health and environmental impacts or, indeed, their LP-mitigation capabilities (Shaw, 2014). Thus, while LED's exponential growth risks nullifying the energy gains of the lighting sector, its potential for eco-friendly lighting practice and thus LP mitigation tends sadly to be under- or un-exploited (Schulte-Romer *et al*., 2019).

To summarize, adverse impacts of artificial nighttime lighting stem from excess illumination outside target areas—especially uplight—and at times when lighting may be safely dimmed or switched off, nuisance glare due to the visibility of the light source causing temporary blindness or discomfort, and the presence of wavelengths shorter than the orange part of the spectrum, or about 585 nanometers.

Rationales for planners' involvement in LP-mitigation

Principal rationales for planning include to prevent and mitigate negative externalities of economic activities and to support the provision of public goods (Klosterman, 1985). The previous section's discussion about the human, ecological, and social impacts of LP, LP's threats to humanity's shared cultural heritage in the view of the natural dark sky, and the need for collective action for meaningful impact, all thus underscore the need for planners' attention to LP. Additionally, LP regulation falls within the ambit of local planning: urban land uses that contribute to community life and vitality—such as manufacturing, trade, transportation, and recreation—are also key LP sources (Olsen *et al*, 2014). Their growth is influenced by local planning, and zoning codes already contain standards requiring lighting and specifying related parameters (Ploetz, 2002). Relatedly, municipalities are ideally located to enforce pollution laws because they have incentive to act, knowledge about sources that present the greatest threat, flexibility to pursue violators, and the ability to respond quickly (Lehner, 1993). Third, while the

costs of wasted light and related carbon impacts indeed are informing policies, they do so in incomplete ways and with contradictory results. This sets the stage for public and stakeholder engagement in decisions in which planners routinely facilitate information-sharing across boundaries, deliberation about goals and strategies, negotiation, and consensus-building (Innes and Booher, 2010; Forester, 2006; Burby, 2003). Significant knowledge about LP's adverse impacts as well as desired lighting qualities for LP mitigation is already available, as is related technology, all of which should ease related planning collaborations with professionals in public lighting, utilities, design, and environmental conservation. Local planners are thus well situated to help shape related policies. Interest in the topic appears to be growing: the American Planning Association's Planning Advisory Service received more than 400 lighting-related inquiries in the previous decade, according to a 2004 *Zoning Review* article (Crawford, 2004, p. 3).

A final rationale relates to planners' ethical obligations in light of LP's ecological and disparate social impacts. The benefits and costs of excessive artificial light are distributed unevenly with many of the benefits—from advertising, extension of shopping hours, nighttime entertainment, etc.—going to business owners, while many costs, such as adverse health and ecological impacts and the loss of the night sky, are borne by the public as a whole and nature and wildlife. Planners are exhorted to pay special attention to the needs and interests of less powerful groups, future generations, and non-human actors, and to preserve the natural environment (APA-AICP, 2016). As urbanization spreads across the globe, planners are urged to conserve nature and wildlife in the urban communities in which they work (Beatley, 2011).

Exploring light pollution regulation in eight cities: Methods

How do localities regulate LP, if they do so at all? What rationales, purposes, goals, and adverse impacts among those discussed above (or others) guide policies? What is the content of policies, and how do they vary based on land use or purpose, such as street lighting, sports arenas, or signs? How do they balance competing objectives such as safety, advertising, energy-efficiency, and biodiversity? What, if any, are exemplary approaches to LP mitigation? How do planning and public lighting professionals and elected officials view LP and community-level factors that might support or hinder related policy-making and implementation? Finally, what recommendations emerge for planning practice and scholarship?

To answer these questions, I launched in 2020 a qualitative mixed-method study in which specific questions and methods emerged incrementally as the study unfolded. As an initial step, I reviewed zoning codes, comprehensive plans, and, if available on city websites, other policies such as those related to environment, streets, and design—in eight cities in the United States. I selected two of the largest cities—one northern and one southern—from each of the four major ecological zones dividing the continental U.S. from east to west. See Table 1. I was interested in larger cities distributed across the country, ones that likely would have the full range of land uses implicated in excess lighting, as well as municipal capacity to develop and enforce LP-related policies. The ecological zones delineate key bird-migration corridors or flyways, thereby offering an ecological parameter for case-selection (IDA, n.d.-c). Birds exist within everyday urban landscapes and migratory birds are dramatically affected by LP. Northern and southern cities represent divergent seasonal characteristics and temperature ranges that have implications for outdoor activity at night. I was also interested in exploring the state of LP-related policies in "regular" cities, that is, those that lack the compelling rationales that typically drive local LP-control, such as the presence of major astronomical observatories, national parks, or the

protection of endangered species. The selected list includes some of the country's largest cities, all with universities nearby as well as resources in environmental and technological fields.

Table 1 about here.

Zoning and other codes are of special interest in the study because they contain standards that are legally enforceable. To isolate terms pertaining directly or indirectly to LP, a research assistant and I searched for keywords such as light, light pollution, illumination, luminaire, glare, sky glow, light trespass, uplight, spill-light, lumens, LED, and others derived from a review of policy-oriented research on light pollution first in zoning codes, then expanding to other codes such as those related to utilities, signs, and streets. Finding only minimal evidence of LP-related concerns, we searched on city websites for master plans, neighborhood plans, and other plans and reports. Within relevant sections, we independently coded explicit and implicit rationales for requiring, encouraging, or restricting lighting, and relevant standards such as lighting provision and operation, minimum and maximum illumination levels, curfew for dimming or switch-offs, location, fixture and light characteristics, and other pertinent features. I also analyzed the lighting code of Flagstaff, Arizona, widely considered best-practice with documented mitigation impact.

After assembling LP-related findings (or lack thereof), I reached out by email and telephone to planners in each study city to obtain reactions to findings, contextual information and perspective on specific LP measures or their absence, and possible motivators for change in the community. I interviewed a total of 12 individuals—local planners, lighting agency staff, a local expert, and an elected official—from six out of the eight sample cities and Flagstaff, with non-response, despite several attempts, from New York and Nashville. Interviews ranged from about 15 minutes to nearly an hour. Finally, we scoured news media sources in, or pertinent to, each city to gauge the nature of LP-related concerns, actions, and controversies, if any existed,

over the last decade. Content derived from interviews and media reports was coded for responses to the study's questions and also to identify relevant unanticipated or emerging themes. Note that the main objective of this exploratory study was to sketch the state of practice related to LP-control and mitigation in some of the country's largest cities rather than to investigate lighting standards *per se* or to compare cities across specific categories.

Findings and discussion

Lighting provision and regulation in municipal codes

Expectedly, zoning codes of study cities commonly reference such longstanding rationales for outdoor lighting as promoting public safety, safe and comfortable navigation for pedestrians and other travelers, business and entertainment, sports and recreation, and aesthetic character. See Table 2. Accordingly, standards require the provision of light in various districts, corridors, and activity areas, and specify minimum illumination levels, duration and directionality (to minimize light trespass into neighboring, especially residential, properties). They specify fixture characteristics such as pole heights and spacing intervals; require that lighting must not conflict with federal aviation rules or be confused for traffic signals; and may detail luminaire elements such as light source, shielding, and lens-type. Less typically, they may specify light characteristics such as coordinated-color temperature (CCT) and color-rendering index (CRI)[3] Additionally, codes require lighting schemes to be compatible with the existing pattern and function of the area or to preserve historic or aesthetic character. Expectedly, more permissive standards for illumination occur in industrial, commercial, entertainment districts; some even have special sign districts.

Table 2 about here.

In addition to such light-affirming rationales and standards, all codes also regulate to prevent or minimize glare, nuisance trespass affecting adjacent properties—especially residences—and public streets, and distraction to drivers posed by dynamically illuminated billboards. Standards to prevent glare and trespass typically call for shielded luminaires, though seldom with specificity on desired performance. For example, Houston requires "shades or hoods on fixtures" to prevent light from shining into residential properties and Jacksonville calls for the design and operation of lighting systems to minimize glare, shadow, light pollution and light trespass. An exception is New York's requirement that lights in public areas abutting residential uses have semi-cut-off shielding such that no more than 5% of peak illumination radiates above 90 degrees and no more than 20% above 80 degrees of nadir (the lowest point of the lamp).

A couple codes also specify a 10 PM curfew for playfields (Nashville and Seattle) with Los Angeles additionally requiring automatic timer control with on-off switching for recreation areas and prohibiting the use of floodlights at construction sites after 6 PM. Most codes also specify *maximum* levels of trespass illumination, such as 0.5 foot candle, crossing into residential areas or 1 foot candle into non-residential areas (Nashville); or 0.01 foot candle at 15 feet from the boundary of the target property (Los Angeles). A few places give property owners (Jacksonville, FL) or municipal officials (Los Angeles) discretion to lower lighting levels from those specified in the zoning code.

Seattle's code is unique among study cities to cite a *preference* for zero lighting in waterfront areas: "Artificial night lighting shall first be avoided." If that is infeasible, the code urges "minimal impacts on the aquatic environment," including with the use of shades and by focusing light on the pier surface. Certain waterfront recreation areas are required to be without artificial illumination altogether. More typically, codes urge restraint in waterfront areas and

scenic corridors in general terms. Los Angeles stipulates street lighting on scenic corridors to be kept to "minimum useful intensity" and located only at intersections and parking sites and away from the roadway. Without detail, the city's code also instructs that emission of artificial light from any seaside industrial/ commercial facility be controlled. Thus, codes' curbs on adverse impacts of lighting on the natural environment are general exhortations rather than specific performance standards. On the other hand, the specific measures identifying trespass, such as the 0.5 or 1.0 foot candle noted above, are without scientific basis, according to a lighting expert I interviewed.[4]

Sign-related lighting codes

As the billboard industry increasingly incorporates LED technologies with dynamic effects, color changes, and videos, concerns related to glare and trespass onto residential properties and distraction to roadway users are heightened. Standards acknowledge commercial interests while also seeking to minimize adverse impacts on residential properties, users of streets and outdoor spaces, and the aesthetic character of the area. Sign-related codes typically specify particular eligible districts or corridors where illuminated signs may be installed; minimum distance of such signs from residences, streets, and intersections; directionality, character, and operation of lighting; limits on dimensions of illuminated portion; and less often, curfews related to extinguishing or dimming of lighting.

To illustrate, codes in Jacksonville, Nashville and Omaha prohibit sign lighting from causing glare or otherwise impairing vision. Omaha's specifies a maximum illumination level of 5200 nits[5] during daylight hours or 0.3 foot candle higher than ambient level, whichever is less; and between dusk and dawn, a maximum level of 500 nits or 0.3 foot candle higher than ambient level, whichever is less. Houston allows a generous maximum of 1250 nits between dusk and

dawn for certain dynamic signs, and 3500 nits for Special Facility Display Signs. Study cities typically disallow signs with flashing lights in all or most districts, making exceptions only for specific areas such as convention, entertainment, and arena districts.

The above discussion showcases the affirmative nature of outdoor lighting standards: to facilitate night-time travel and outdoor comfort, business, entertainment, and other activities while also promoting public safety and protecting residences from nuisance glare and light trespass. Rationales related to broader public health, environmental conservation, LP-control, and dark-sky concerns are typically absent or written as preferences rather than standards. The vast majority lack standards specifying fully-shielded luminaires (allowing zero light above the horizontal plane), light sources, or spectral qualities of illumination. No code limits the cumulative density of lighting in any land use district, activity area, or property, and none seeks justification for new or additional lighting.

LP mitigation in other policies, local interviews, and media reports

Beyond zoning and other relevant ordinances, content analysis of comprehensive, area, and sectoral plans (such as for environment or transportation) and policies revealed a mix of results (See Table 3). Documents from most cities were silent on LP issues or repeated the glare and light trespass concerns of zoning codes discussed previously. Policies from Chicago, Los Angeles, and Seattle contained mention of light pollution, including the need to direct light away from wetlands (Seattle) and protect birds (Chicago). The Framework Element of Los Angeles' Comprehensive Plan identifies a goal to "protect and preserve the nighttime environment … and otherwise minimize or prevent light pollution…" (Goal 9P). We also found reference to 'sky glow' and a goal related to "a street-lighting system that protects and preserves the nighttime

environment … with minimum impact on the environment …" in the Granada Hills-Knollwood Community Plan in Los Angeles (CF 15).

Omaha's Master Plan has the greatest coverage of light pollution issues of this study in its Environment element. For example, the Visual Resources section (p.22) states that lighting should "minimize adverse impacts of … light pollution on … night-sky views," with recommendations to establish standards that minimize impacts on wildlife, encourage reduction in the operation of lights that may adversely affect wildlife (p.22), and collaborate with professionals in evaluating impacts on wildlife, particularly migrating birds (p.27). The Building Sites section recommends the development of a "'dark sky' lighting ordinance that directs light away from the sky" and possible "increased limits on the amount of light that is allowed to cross property lines" (p.85).

In response to Chicago's 2006 Nature and Wildlife Plan containing a goal to reduce "negative effects of Chicago's ambient light on nocturnal animals and insects," an Update (2011-2016) reports that a public-private collaboration expanded the "Lights Out!" program to protect migrating birds from crashing into illuminated buildings, whereas "no specific programs have been formulated to address other animals and insects." (pp. 17, 30). The Chicago River Design Guidelines Checklist contains elements related to lighting including color temperature of LED bulbs (white light with color temperature of 3000K or below) and light pollution ("dark sky compliant" fixtures) (pp 73, 80). Lamenting the merely advisory status of the guidelines, the Chicago planner added, "how we treat lighting doesn't do enough" for birds and other wildlife.

Table 3 about here.

Interviews with other key informants similarly confirmed and contextualized findings from the code and policy review. Lighting-affirmative rationales such as pedestrian and traffic

safety, navigation comfort, and effective functioning of various districts and activities were reinforced in interviews. Planners communicated the tension between public perception that more light is better and the imperative to support divergent objectives including safety and aesthetics. Emphasizing pedestrian and bike safety especially during winter, for example, as well as the associated risk of liability, our Seattle planner discussed the city's decision not to dim LED streetlights later at night, despite the availability of such adaptive controls. On the other hand, complaints from drivers are driving calls for stricter regulation of illuminated billboards, as interviews of planners in Los Angeles and Omaha revealed.

Interviews also highlighted specific restrictions that incidentally or unintentionally reduce light pollution. For example, Jacksonville's Whitehouse naval training airfield requires strict optical controls on nearby outdoor lights so as to simulate mid-ocean night landings on ships, a finding confirmed in the city's 2030 Comprehensive Plan. In another example, California's Proposition 218 requires cities to conduct ballot initiatives asking residents to vote to assess themselves for the costs of streetlight maintenance. Lighting districts in which a majority of residents reject the assessment have their lights turned off, effectively creating darkness in those districts. Our LA respondent also pointed to the absence of specific LP-related activism or grassroots clamor to control LP, and thus of political champions, as a key reason for lack of intentional LP control, a common explanation for the lack of attention to LP across cities.

Albeit less frequently, respondents described technical challenges to implement LP mitigation. For example, our Seattle respondent discussed efforts to change luminaires following community complaints related to the earlier 4000K LED installations. While they are voluntarily changing luminaires to 3000K or less, such fixtures fall short of some SDOT requirements for arterials for illumination levels, uniformity, and efficiency given current pole spacing. Because

streetlights are funded from the city's general fund, and given LED's longevity, upgrading is done only as current lights fail. Spending moratoriums on streetlight conversions during the Covid pandemic have created additional delays.

In diverse cities, LED streetlight conversions dominate news media reports related to outdoor lighting over the last decade. Such reports credit energy efficiency as the driver with claims of environmental—meaning CO_2 reduction—as well as budgetary benefits. Gavin (2021) reports that the Chicago Smart Lighting Program's conversion of 270,000 HPS lamps to LED translates into 50% of energy-savings over 10 years, equivalent to approximately $100 million. The article also cites an industry survey finding that 185 of the 300-plus largest US cities are pursuing LED street-lighting, representing a market that ranges from 600,000 to 1.5 million units per year and a 35% projected penetration of smart streetlights in the country by 2030. Even though energy and cost savings arguments pervade media reports, concerns also exist related to the disruptive effects of excess lighting resulting from such conversions in Chicago (Fruend 2020; Stark 2018) and Houston (Holeywell 2017; Shelton 2016); negative effects of the blue-heavy LED lighting including on migrating birds in New York City (Gibberd 2006; Norman 2019); and on amateur astronomy in Nashville (Hughes 2018, Gonzalez 2011) and, as well, concerns related to pedestrian deaths due to *inadequate* street lighting in Nashville (Gonzalez 2021, Miller 2022).

Despite LED's promise of cost-savings due to energy efficiency and longer life-span, initial set up costs are high. Increasingly, energy service companies (ESCOs) are funding conversion projects which help cities avoid such up-front costs and, at least on paper, realize significant savings over a number of years (Gavin 2021). Jacksonville's streetlight conversion program, however, raised concerns about even the cost-savings argument. To illustrate,

Stepzinski (2016) reported that the city claimed savings of at least $1 million annually in electricity costs after full replacement. However, data obtained from the city in 2020 showed that the program ended up costing nearly $35 million, a 45% increase over the 2014 projection of $24 million (Becker, 2021). Furthermore, the LED lights that replaced the high-pressure sodium ones have a color temperature of 4000K, much higher than the 3000K maximum recommended by the American Medical Association for streetlights.

Enacting LP control can be politically challenging. Responding to opposition from the business community, Governor Pataki of New York refused to sign a bill that would require local government action to advance dark sky objectives (McKinley 2002). For more than a decade afterwards, New York city councilmembers repeatedly introduced amendments to the city's codes, without success, to require that streetlights and other external lights be fully shielded and have a CCT of no more than 3000K (See Table 3). This analysis thus reveals a pressing conundrum for advocates of LP control: even as advocates themselves center, entirely legitimately, energy-efficiency and costs-savings in their case for regulating LP, LED conversions and higher lighting levels are, nevertheless, resulting in increasing LP, along with their varied human and ecological impacts. At the same time, powerful stakeholders challenge local LP control policies, as seen in New York City. Thus, even increasing awareness of the need for urgent climate action has yet to yield the nuanced understanding of potential conflicts in objectives related to carbon mitigation and biodiversity protection in LED conversions or, equally significantly, of the possibilities for achieving both (and other values) in the specification of LED lamps and related adaptive controls. And where both, knowledge and willingness to change, exist, technical hurdles, financial constraints, or the fear of liability can delay or stymie

desired actions such as selecting lamps that are safer for wildlife or dimming or turning off lights later at night.

What specific actions might planners propose, then, to obtain meaningful impact, at the very least, in reducing the rate of LP growth in cities, if not achieving more ambitious objectives? Planners might study cities that have adopted specific LP-related standards and compliance protocols and that track impacts. Planners might also peruse "model ordinances" developed over the last couple decades by the International Dark Sky Association (IDA, n.d.-d; IDA-IES 2011; IDA 2018; Crawford, 2004). Several papers offer surveys of and or planning advice on local lighting codes (Green *et al*, 2022; Luginbuhl *et al*, 2009; Ploetz, 2002; Crawford, 2004; Jewkes 1998); one study, however, documents positive impact of the codes (Luginbuhl *et al*, 2009, discussed in the next section). It is beyond the scope of this paper to offer a detailed discussion of IDA's model ordinances. Briefly, however, IDA-IES's Model Ordinance (2018) has been criticized by several authors, including for permitting more light in many districts resulting in sky glow impacts that are greater than what can be expected even when lighting is unregulated (Illinois Coalition for Responsible Outdoor Lighting, 2011; Morgan-Taylor, 2015; Luginbuhl, 2013). IDA's (2018) Certified Dark Sky Community (CDSC) model ordinance has been adopted by 17 U.S. communities (IDA, n.d.-d). Among other standards, the ordinance requires full shielding of lamps over 1000 lumens, and limits short-wavelength light and the total density of illumination, all helpful guidance for planners. The CDSC certification, however, involves self-identified communities—typically smaller towns in mostly non-urbanized regions—that display "exceptional commitment to and success in" pursuing dark sky preservation and put into place several measures to ensure continuing certification (IDA, 2018).

Flagstaff, Arizona's Lighting Code

Few studies document the impact of lighting codes. Luginbuhl *et al* (2009) is a notable exception for their investigation of Flagstaff, Arizona, whose standards are widely considered to be a model for achieving dark sky objectives, even if they are far from perfect. The study finds a 17% growth in light escaping into the sky from 1989 to 2003, compared to an expected 43% increase had the city's lighting code that limits total outdoor lighting on new construction not been enacted. Additionally, if all legally nonconforming lighting installed before 1989 were brought into compliance with the code, sky glow in Flagstaff could decrease by 36% compared to that in 2003; moreover, if all lighting, including residential, could be converted to *fully* shielded fixtures, sky glow would decrease to about half the current value.

Analyzing both the specific (per acre or per unit) and total amounts of light used for different land uses, the study documents sports lighting as the largest user of lighting or 32% of the city's total when it is on (pp. 197-198). Apart from sports lighting, the most intense users are commercial establishments—such as gas stations, restaurants, shopping centers, and hotels—accounting for 36% of total output when sports lighting is all turned on and 53% when it is off. Roadway and residential lighting account for 12% and 14% respectively, with the remaining 21% arising from industrial and institutional uses and automobiles (p. 198). Significantly, unlike most other cities, Flagstaff's roadway lighting is nearly universally fully-shielded, contributing to its relatively lower portion of the total output. Thus, data obtained on lighting installed before and after the adoption of the lighting code helps convey both how Flagstaff might have looked like today without the code as well as possible gains if all lighting were brought into conformance with present standards and if all outdoor lighting were perfectly shielded to avoid diffusion outside of target areas.

Briefly, the lighting element of Flagstaff's Zoning Code contains three principal *objective* standards to limit the cumulative adverse impacts of lighting. See Table 4 for additional details.

Table 4 about here.

a) *Limits on the total amount of light used per acre*. Lumens per acre caps produced 65%-plus reduction in uplight output for new developments relative to the level expected if the caps had not been adopted in 1989 (Luginbuhl *et al*, 2009, p. 201). The 2022 amended lighting code identifies two lighting overlay zones, Lighting Zones (LZs) 1 and 2, specified with reference to proximity to the US Naval Observatory, with nearer areas permitted less output per acre. Each zone has specific standards related to maximum output, shielding, and curfew for different land uses within each zone.

For example, the code allows a maximum of 17,500 lumens per acre for commercial, industrial, and multi-family uses in LZ1, and 35,000 lumens per acre for these uses in LZ2. For single family or duplex residential lots, the code allows a maximum of 1,000 lumens per lot or parcel in LZ1 and 5,000 lumens per lot or parcel in LZ2. Because fixtures attached to buildings—especially on walls with low reflectance qualities—and mounted under roof overhangs or canopies contribute less ambient light than free-standing ones distant from buildings, less output from the latter types is allowed in LZ1.

Notably, these lumen counts are lower than the previous version's 25,000 and 50,000 lumens per acre for LZs 1 and 2, respectively, for industrial, commercial, and multi-family uses, and 10,000 lumens per parcel for single family and duplex lots in both zones. Flagstaff-based respondents contextualize these changes as a response to the increasing use of more efficient LED technology and the phasing out of LPS lights for which the previous standards incorporated a 0.7 multiplier to measure actual output. The updated code also reduces lighting zones from

three to only two. These changes were among six key recommendations from a study of the impact of current and projected lighting trends on the continued operation of the Naval Observatory (Coconino County, 2019).

b) *Full- or near-full shielding of luminaires*. The city's lighting code defines three classes of lighting for residential and non-residential outdoor areas: Class 1, where color rendition is important, such as for outdoor eating, sales areas such as gas station canopies, and primary entry/exit areas; Class 2, such as general roadway and parking lighting; and Class 3, or decorative lighting such as that used for illuminating buildings, flags and landmarks, or landscape elements. For each class, the code specifies zones where fully-shielded lighting is required, where it is preferred even if partially-shielded lighting is permitted, and where lamps above a certain lumen output are prohibited altogether.

The code requires fully-shielded fixtures for Class 1 and 2 lighting in commercial, industrial, and multi-family uses for LZs 1 and 2. For these uses, Class 3 lamps with an output below 1,750 lumens are required to be fully shielded in LZ1 while they may be partly shielded in LZ2; lamps at and over 1,750 lumens are prohibited in both zones. For single-family and duplex residential lots, lamps of all Classes with an output of less than 700 lumens are required to be fully shielded in LZ1 and permitted to be partially shielded in LZ2; lamps at and over that count are prohibited in both zones. As with the lumens per acre standard, these standards too represent a reduction of allowable lumen-output per fixture from the previous version to adjust for LED's greater lighting efficiency.

c) *Lamp standards*. Additionally, the 2022 Lighting Code requires all Class 1 lighting to have a CCT of 2700K or less, and Narrow-Spectrum Amber LED lighting for all Class 2 lighting applications. The updated standards also restrict the spectral range of the light source relative to

the older Code, specifying peak wavelength to be between 589 and 595 nanometers. (The previous allowable range was 585 to 595 nanometers.) All light sources may be used for Class 3 lighting, including those incorporating color changes. While all LED lights contain blue wavelengths, narrow-band Amber LED lighting reduces the content of blue and higher-energy wavelengths while also offering energy, astronomical, and visual sky glow advantages.

As for signs, except for exposed neon lighting, the Code also treats external illumination for signs as Class 1 lighting, counts it toward the total outdoor light output limits, and requires that external lighting be steady, stationary, downward directed, and fully shielded. The Code also distinguishes standards for displays of letters and symbols on signs in Lighting Zones 1 and 2 and imposes a time limitation of 9 PM in LZ1, and 11 PM or when the business closes, whichever is later, in LZ2.

Thus, Flagstaff's Lighting Code offers objective standards organized by land use, land area, Lighting Zone designation, and fixture type. As before, all lighting owned and operated by public agencies is required to be fully shielded. Applicable to new construction and substantial rehabilitation or replacement or change in use, the standards nevertheless stand in stark contrast to those related to the cities reviewed in the first section. Thus, standards operationalize the Code's purpose as outlined in its opening: to "help assure that dark skies remain a resource to be enjoyed by the Flagstaff community and its visitors, minimize light pollution and light trespass across property lines, maintain an environment that the City's astronomical industry depends upon, reduce energy consumption, minimize the potential disruption to nocturnal ecosystems, and to allow safe and efficient outdoor lighting."

Flagstaff's planner explained the lighting code's development and its recent updates. As LPS sources were phased out in the marketplace and LED usage became ubiquitous, the city saw

wholesale replacements using LED fixtures especially on commercial properties, often without the requisite permits. Following a Joint Land Use Study (Coconino County, 2019) conducted in partnership with the county, nearby military installations and other stakeholders, to assess the impact of existing and projected illumination, standards were modified to apply the stricter LZ2 standards to LZ 3, revise total lumen output per acre and per fixture, and restrict further the allowable spectrum for Narrow-Spectrum Amber LEDs. “We work with property owners and contractors all the time. For commercial owners, there’s a lot of hand-holding ... For example, in an old mall, we worked to bring them in compliance by showing how they could phase their replacements,” our contact noted.

In keeping with its status as the world’s first Certified Dark-Sky Community, the city dedicates funds to support enforcement, education, and outreach to encourage voluntary compliance and track non-complying fixtures and properties, and bolstered these with the purchase of a spectrometer and related technology. The lighting enforcement office is also responsible for assembling annual Dark Sky reports posted to the city’s website. Professional astronomers affiliated with local observatories offered a great deal of expertise over the years in the development of the code, trained staff and elected officials, developed educational materials, participated in media and community outreach, and advised studies. “It took a lot of work to become the world’s first certified first Dark Sky Community,” our planner noted, however, “it is not as hard as it is sometimes made out to be. It is possible. It is feasible. We’re the example.” Offering other levers for communities without significant astronomical activity, such as arguments related to LED’s impacts on circadian rhythms and nature, she stressed the need to make informed tradeoffs to balance multiple values: “White LED *is* more efficient than narrow-band Amber LED, but a little more electricity is worth the lower impact on dark sky and

nocturnal wildlife." Flagstaff thus owes its success to efforts over more than sixty years to control sky glow, with ongoing technical assistance from astronomers, outreach to residents and policy makers, and support from state and federal government, all in the face of the rapid embrace of more light-polluting sources and cheaper technology, as well as community growth.

This study has a few limitations. A first planning study of its kind, it examined zoning codes in a relatively small sample of cities for LP regulation. As such, it was designed to offer a bird's eye view of the state of practice, not to compare cities' codes or performance with each other. Beyond zoning and other ordinances, it investigated communities' goals and policies related to LP only for those cities for which such documents were readily available. A more thorough search might reveal greater desire and perhaps even intention to regulate LP than is evident from the documents reviewed. Finally, it is possible that additional political or technical challenges (or opportunities) might surface in more in depth case studies of individual cities with wider participation of representatives. These and other gaps could be filled in future scholarship.

Conclusion

Zoning and other ordinances in some of the major cities across the country address outdoor artificial lighting predominantly in terms of its historic role in promoting night time navigation, safety, advertising, and recreation, with restrictions typically designed to avoid nuisance caused by glare and spillover beyond property boundaries. However, even such restrictions are plagued by lack of definition, merely encouraging language, or arbitrary measures. It is also unclear to what extent standards are enforced beyond permit review. A small smattering of master plans, area and sectoral plans, and agency reports are starting to

acknowledge LP impacts on the environment, wildlife, and dark sky, albeit inconsistently across policies in the same city and without enforceable standards.

Still, existing codes that already require shielding of lamps, directionality of light, maximum output per fixture, and curfews for full switch-off or dimming, might be reviewed against community values and goals, specified more precisely and consistently and with regard to newer technologies, and combined for cumulative impact, all without eroding functional requirements related to safety and comfort. By requiring *fully-shielded* outdoor fixtures especially for streets, parking areas, and billboards, for example; luminaires with CCT of 2700K or less; and 9 PM or 11 PM curfews depending on land use, it is possible to limit increases in light pollution through zoning codes. Conversions of street lights and other public lighting offer a starting point for implementation and related public outreach and engagement.

Beyond sharpening existing standards, cities might explore how lesser-known LP-controls may be phased in, such as limits on: a) total illumination output per acre for different land uses; b) output per luminaire based on purpose, shielding, attachment to and distance from buildings, and other parameters; and c) wavelengths of light in the blue-violet range. Technology exists to adopt the above parameters but modifying wholesale the existing layout of light poles and fixtures to achieve desired qualities of illumination nevertheless will involve costs. Thus, upgrades might be made to Narrow-Spectrum Amber (NSA) LED sources for general purpose lighting, as existing lamps fail. A more 'middle-range' recommendation for streets and parking is Phosphor-Converted Amber LED lamps which are twice as efficient as the NSA LED ones and have more blue and green wavelengths than are desired for dark sky impact, but are superior to the more commonly installed White LED lamps.

Given the somewhat technical nature of these recommendations, planners will need to educate themselves about the growing impacts of light pollution including due to the widespread adoption of LEDs, and specific features of outdoor lighting that can contribute to as well as potentially mitigate LP. They might engage elected officials, stakeholders, and residents in deliberations to clarify values and goals with respect to lighting and translate these into desired standards and benchmarks for implementation, drawing knowledge as well as inspiration from cities like Flagstaff. They might review plans and policies related to land use, environment and natural resources, transportation, design, and others, to ensure consistent treatment affecting outdoor lighting. They will also need to help leaders grapple with the tradeoffs associated with cost-savings versus optimally combining objectives in human health, biodiversity, and carbon mitigation, and in negotiating an uneven arena consisting of lighting promoters such as manufacturers, trade associations, and businesses on the one hand, and naturalists, astronomers, and other LP-control advocates on the other.

University courses in city planning and landscape and architectural design might incorporate LP's human, social, and environmental impacts and strategies for mitigating LP in existing settlements and minimizing it in new developments. Such material might be appended to syllabi related to environmental and natural resource conservation, biophilic urban design, urban infrastructure (including related finance), sustainability (broadly defined, that is, going beyond energy efficiency), and cultural values and assets. As NYC's experience shows, LP mitigation encompasses the conflicts and tradeoffs illustrated in Campbell's (2016) sustainability triangle, which also might merit exploration in the classroom.

As cities are called to address climate shocks, biodiversity loss, and social wellbeing and equity, they undoubtedly will benefit from the creation of federal standards related to light

pollution such as those currently present for air and noise pollution. Leading professional organizations—such as International Dark-Sky Association, American Medical Association, and perhaps even the American Planning Association—might also offer unambiguous, ongoing guidance to help communities achieve desired impacts through zoning and building codes in ways that incorporate emerging technology. In combatting LP, it will be important for advocates and planners to communicate early and often the ways in which more light is far from benign in its impacts; that "environmentally-sensitive" means more than "energy-efficient"; and that tackling light pollution is possible without sacrificing safety or functionality of outdoor spaces.

Acronyms

AMA American Medical Association

APA American Planning Association

CCT Correlated Color Temperature

CRI Color Rendition Index

EPA Environmental Protection Agency

ESCO Energy Service Companies

f.c. foot candle

HPS High-Pressure Sodium

IDA International Dark-Sky Association

LED Light-Emitting Diode

LP Light Pollution

LPS Low-Pressure Sodium

LZ Lighting Zone

NSA Narrow-Spectrum Amber

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Tables

Table 1. Study cities

	Population 2020 Census	Median household income (in 2021 dollars), 2017-2021	All employer firms, 2017	Density: Population per sq. mile, 2020	Land area in sq. miles, 2020	Code Title and Source
New York city, NY	8,804,190	$70,663	205,592	29,303.2	300.45	Zoning Resolution, Articles I-XIV, Appendices https://zr.planning.nyc.gov/
Jacksonville city, FL	949,611	$58,263	18,475	1,270.7	747.29	Code of Ordinances, several titles https://library.municode.com/fl/jacksonville/codes/code_of_ordinances
Chicago city, IL	2,746,388	$65,781	50,497	12,059.8	227.73	Municipal Code of Chicago, several titles; Chicago Zoning Ordinance and Land Use Ordinance https://codelibrary.amlegal.com/codes/chicago/latest/chicago_il/0-0-0-2653760
Nashville-Davidson (balance), TN	689,447	$65,565	13,855	1,449.1	475.78	The Code of the Metropolitan Government of Nashville and Davidson County, Tennessee, several titles, including Title 17-Zoning Code https://library.municode.com/tn/metro_government_of_nashville_and_davidson_county/codes/code_of_ordinances?nodeId=CD_TIT17ZO
Omaha city, NB	486,051	$65,359	11,266	3,433.4	141.57	Code of Ordinances, several chapters including Chapter 55-Zoning and Chapter 44-Electricity https://library.municode.com/ne/omaha/codes/code_of_ordinances
Houston city, TX	2,304,580	$56,019	55,378	3,598.4	640.44	Code of Ordinances, several chapters including Chap. 33-Planning & Development and 32-Parks and Recreation https://library.municode.com/tx/houston/codes/code_of_ordinances
Seattle city, WA	737,015	$105,391	22,311	8,791.8	83.83	Code of Ordinances, including Title 23-Land Use Code and Title 25-Environmental Protection and Historic Preservation

						https://library.municode.com/wa/seattle/codes/municipal_code
Los Angeles city, CA	3,898,747	$69,778	99,177	8,304.2	469.49	Municipal Code, several titles including Title 22-Planning and Zoning and Title 27-Electrical Code https://library.municode.com/ca/los_angeles_county/codes/code_of_ordinances

Source: US Census Quick Facts. Sources for codes as noted for each city.

Table 2: Zoning code purposes and concepts related to outdoor light at night

Purposes, goals of light at night	**Concepts**
Public safety, comfort, welfare • Pedestrian and vehicular safety, navigation • Crime prevention • Protecting residences from nuisance glare or spill-light	• Requirement of lighting provision, operation, maintenance • Making the impairment or obstruction of lighting to be unlawful • Minimizing glare and spillover into residential properties, streets, and parks • Fixture aspects including shielding of lamp, and height of pole, location and spacing • Coordination of lighting layout with planting, landscaping • Required hours of lighting between dusk and dawn (or sunset and sunrise) • No lighting that may be confused for traffic signals, no conflicts with federal aviation rules **Less common** In waterfront and other public areas (NYC) • Minimum lighting levels, and average-to-minimum ratio of lighting • Specification of CCT and CRI of lamps in particular uses • Specification of shielding of lamps
Aesthetics • Design or neighborhood character • Compatibility, consistency	• New lighting scheme to be compatible with existing pattern, and design/ function of abutting properties • Common lighting plan for property with more than one business • Preserve historic character in lighting choices • Consider pedestrian interest, safety, comfort • Preserve, protect, enhance character of entertainment district (NYC) • Scenic area with high mast lighting not to be designated scenic (Houston)

Advertising signs, information • Illuminated advertising and billboards • Public message signs	• Specification of location, number, square footage, height, illumination level and other characteristics of lighting • Hours of illumination of signs and exemptions • Protections for nearby residential uses, streets, and specific areas from glare and light trespass • Specific standards related to dynamic, self-illuminated signs • Specification of special sign districts, with required minimum lighting level and duration (e.g., Times Square in NYC)
Energy conservation: • Energy efficiency • Green buildings	• Zoning code refers to Energy Code for maximum permissible lighting levels (Seattle) • Zoning code refers to the CA Green Building Code (Los Angeles) • Shielded fixtures • Automatic switch off or dimming during inactive periods between dusk and dawn
Natural resource, heritage conservation: • Protection/conservation of natural areas • Heritage landscapes	• Scenic corridors (Los Angeles): • Restrict light intensity to minimum useful level, • Restricting lighting to intersections and parking, • Locating lighting away from roadway • Control emission of artificial light from any seaside industrial/commercial facility (Los Angeles) • Shoreline areas (Seattle) • Zero lighting preferred • No artificial lighting on natural athletic fields in shoreline parks • Minimize artificial light impacts on water to extent possible
Government authority and competence	• Roles, authority of city with reference to lighting infrastructure • Definitions of sign-lighting related terms • Permitting/licensing process and requirements

Notes: 1. Codes explicitly or implicitly include other, lower-level objectives within the broader categories of, say, safety and design, such as, respectively, preventing theft and minimizing visual clutter. Thus, this listing is not an exhaustive compilation of all such objectives.

2. Minimizing nuisance associated with glare and spill-light is the most common concept, receiving mentions with reference to a variety of purposes such as protecting residentially-zoned property, safe movement of vehicles, neighborhood character, protecting shoreline and marine environments, and with reference to illuminated signs.

Table 3: Light pollution concepts and standards in eight cities' municipal codes and other policies

City	Policy source	LP concepts, intent[1]	Standards (illustrative only); recommendations/ proposals (f.c.=foot candle)
New York city, NY	Zoning Resolution, Articles I-XIV, Appendices	Glare, light trespass	• lighting shall be shielded from direct view, shielded to minimize any adverse effect on surrounding buildings containing residences (and community facilities or streets) • "Glare shall be controlled to a semi-cutoff standard"[2]
	Planning Together Comprehensive Planning Framework, 2020	NA	NA
	OneNYC 2050, 2019 (full report)	NA	NA
	PlaNYC, Getting Sustainability Done, 2007; PlaNYC Progress Report, 2014	NA	NA
	Proposed Amendments to the New York City Administrative Code; City Council Resolutions (not adopted)	Glare, light trespass, light pollution	<u>Introduced in City Council but not enacted/adopted</u> • Requirement that street lamps and other public lighting have CCT no higher than 3000 K (introduced 2015, 2018, 2022) • Requirement of full shielding of lighting in residential areas not operated by public agencies or for parking/pedestrian areas in excess of 3000 lumens (introduced 2019 and 2022) • Requirement that only full cutoff (fully shielded) fixtures be used for city street lights (introduced 2008, 2011, 2017, 2018) • Requirement that exterior lights be fully shielded when they create undue glare (introduced 2009, 2010) • Requirement that exterior lighting be fully shielded (introduced 2010)

[1] Light-affirmative rationales such as navigation, safety, and aesthetics are not included here; neither, given its near-ubiquity as well as near-irrelevance for reducing ambient light output or sky-glow, is energy-efficiency.

[2] Semi-cutoff standard is defined as "not more than five percent of peak foot candle intensity radiating above 90 degrees and 20% of peak intensity above 80 degrees."

			• Resolution urging New York State Legislature to adopt legislation controlling light pollution, Healthy, Safe, and Energy Efficient Outdoor Lighting Act (introduced, 2008, 2010)
Jacksonville city, FL	Chap. 656-Zoning Code	Glare, light trespass	(For four of five "Character Areas"): "Provide lighting systems that minimize glare, shadow, light pollution, and light trespass." • Prohibited: All sag lenses, drop lenses, and convex lenses • Illumination levels at property lines not to exceed 0.5 f.c. when located adjacent to residential properties; not to exceed 1.0 f.c. when abutting nonresidential properties. • Required: Use of "cut-off fixtures" with diffusers to minimize or eliminate light trespass • Lamp sources within parking and pedestrian areas to be metal halide, compact fluorescent or LED; LED is preferred
	Chap. 326-Signs and Outdoor Display Structures	Glare, light trespass, distraction	Unlawful: Sign structures with lighting • that produce glare to vehicular traffic or electric incandescent bulbs with a rating exceeding 40% of the lumen output of a one-hundred watt clear bulb, with the lighting located less than 20 ft. above the ground surface • which pulsates, flashes, flickers, alternates or otherwise changes intensity, where the illumination is located within 10 ft. of a street right-of-way.
	City of Jacksonville 2030 Comprehensive Plan, Future Land Use Element	Uplight	2.5. a lighting regulation zone around Outlying Field (OLF) Whitehouse, requiring that all artificial lighting equipment have positive optical control so that no light is emitted above the horizontal plane.
	Chapter 718 – Electric Utilities	NA	NA
	Title X (Chapters 360-387), Environmental Affairs	NA	NA
	Title XXI (Chapters 710-754), Public Works and Utilities	NA	NA
Chicago city, IL	Title 17-Chicago Zoning Ordinance	Glare, light trespass, distraction	17-6-0204-C All lighting to be located, designed, and operated to minimize light spillover and glare onto R-zoned properties • Gas stations. Lighting must be directed downward and shielded

		• Wireless Facilities. Towers may not be artificially lighted unless required by FAA; design to cause "least disturbance" to surrounding views • Parking garages to provide shielding of lighting so as to minimize glare on adjoining properties 17-12-0700 (to -0711) Prohibited signs: strobe lights, video display signs (except in sports stadium and convention center), signs that flash a beam of light into the eyes of a driver upon a right-of-way within 200 ft. from sign 17-15-0505-A Amortization. Flashing signs that become nonconforming must be removed or altered to comply no later than 5 years of the effective date of the amendment
Municipal Code of Chicago**,** Title 9-Vehicles, Traffic and Rail Transportation Title 10-Streets, Public Ways, Parks, Airports and Harbors Title 11-Utilities and Environmental Protection	NA	NA
We Will Chicago Framework Plan (2023)	NA	NA
Comprehensive Plan (1966)	NA	NA
Nature and Wildlife Plan Update (2011-2016)	Dark sky, darkness preservation	P17: Explore ways to reduce the negative effects of Chicago's ambient light on nocturnal animals and insects. P17. In 2006, the "Lights Out" program was expanded to more building levels, hours of day, and participants. P30. "Beyond the efforts for birds, no specific programs have been formulated to address other animals and insects."
Nature and Wildlife Plan, 2006	Dark sky, darkness preservation	P19. Publicize The "Lights Out" program to communicate environmental benefits from turning off decorative exterior lighting and utilizing interior shades; monitor and recognize compliance. • Additionally, review city lighting codes for high-rise construction sites to promote design elements that protect birds
Adding Green to Urban Design, 2008	Excessive ambient light, glare, light	P23. Minimize light pollution, light trespass and glare P35. Develop and adopt outdoor lighting efficiency and design standards, including:

		trespass, light pollution	• P50. Research light trespass regulations in the U.S. • P50. (Possibly) require full cutoff luminaries for all exterior lighting (no horizontal light leaving the site). • P50. Reduce street lighting levels to RP-8[3] standards.
	Chicago River Corridor Development Plan (1999)	NA	NA
	Neighborhood Design Guidelines, 2020 Draft	NA	NA
Nashville-Davidson (balance), TN	Title 17-Zoning Code for Metropolitan Nashville and Davidson County	Glare, light trespass, distraction	• All light and glare shall be directed on-site to ensure surrounding properties are not adversely impacted by increases in direct or indirect ambient lighting levels. • No lights permitted on towers except as required by state or federal law • Religious institution: No illumination in excess of 0.5 f.c. to be permitted across residential boundary or public street; parking lot to be illuminated only when in use • Recreation. No lighting of outdoor amphitheater, ballfields, etc. after 10 p.m., any day of the week • Prohibited: signs with varying intensity, blinking, flashing, etc. movements that convey the illusion of movement; signs that cause glare or impair the vision of motorist or pedestrian on public way
	The NashvilleNext Plan (Comprehensive Plan)	Darkness preservation	Health, Livability, Built Environment Element p145: (Rural character) Infrastructure appropriate for higher population density communities, such as street lighting, should be avoided.
	Nashville Electric System, Street Lighting Guidelines (2021)	NA	NA

[3] These are specified by the American National Standards Institute and the Illumination Engineers Society for roadway and parking lot lighting. The standards offer lighting design criteria to help lighting engineers consider the following goals: Improve motorist visual quality; Provide quality light and increased contrast for seeing hazards; Illuminate conflict areas; Minimize environmental impacts of light at night; Employ lighting systems that are easily maintained and minimize energy use. Read more at ANSI/IES RP-8-21: Design Of Roadway Facility Lighting, https://blog.ansi.org/?p=160622

Omaha city, NB	Chapter 55: Zoning Code, Chapter 44: Electricity	Glare, light trespass, distraction	• Light to be positioned so as not to be directed onto adjoining property or public street • Parking: screening to block the sight line of headlights into residential properties • Electronic signs: no flashing lights allowed; light shall not cause glare or impair vision. • Brightness of sign not to exceed 5,200 NITS during daylight hours, or 0.3 f.c. over ambient lighting, whichever is less. Brightness not to exceed 500 NITS from dusk to dawn or 0.3 f.c. over ambient lighting, whichever is less. • Screen transitions should not appear to flash due to brightness differential between background and content. • Recreation fields: lighting to be extinguished no later than 1 hour after end of event
	Master Plan (2015), Environment Element	Light pollution, light trespass, darkness preservation	**P22. Visual Resources:** Minimize adverse impacts of light pollution on night-sky views, balanced with providing for public safety. P26. 3. Adopt and promote standards for lighting while allowing views of the night sky. 3.1. Establish standards and practices that minimize impacts on wildlife. • Coordinate with professional ecologists. • Encourage reduction in the operation and orientation of lights that may adversely affect wildlife. • Develop lighting information and guidelines for residents. • Work with business districts, … building owners and managers, and school districts to minimize lighting that may result in bird-building collisions and light trespass. 3.2. Develop information and design guidelines to avoid excess lighting and to avoid light trespass on adjacent property. 3.3. Coordinate with public utilities to reduce excess lighting on public property (e.g., right- of-ways). 3.4. Encourage energy conservation by eliminating lighting that is unnecessary **P27. Natural Habitats:** P30. 5. Minimize lights and other disturbances on wildlife in the design of infrastructure and development.

			5.1. Ensure that collaborative efforts are taken when evaluating the impact on wildlife, particularly migrating birds... • Initiate a "Lights Out" campaign to encourage building owners and managers to minimize lighting at night, particularly during bird migration seasons. **P85. Building Sites--Residential and Non-residential.** P88. 9. Minimize off-site environmental impacts associated with light 9.2. Develop a "dark sky," City lighting ordinance that directs light away from the sky. Consider increased limits on the amount of light that is allowed to cross property lines.
	Master Plan, Urban Design Element	Spillover light	P42. Goal 7. Lighting of significant structures. Lighting should be carefully directed to minimize light spill beyond the structure
	Master Plan (Concept, Public Facilities, Housing, Transportation, Land Use, Parks, Urban Development Elements)	NA	NA
	Area Plans: Downtown Omaha 2030 (2010)	Glare, light trespass, distraction	P222. Sign lighting shall minimize light and glare on surrounding rights-of-way and properties. • External light sources to be directed and shielded to illuminate sign only; prevent off-site glare. • Sign lighting shall not blink, flash, flutter, or change light intensity, brightness, or color. • Colored lights shall not be used so as to be confused or construed as traffic control devices. • Neither the direct nor reflected light from primary light sources shall create hazards for pedestrians or operators of motor vehicles. • For energy conservation, light sources shall be fluorescent or other source of equal or greater energy efficiency. Incandescent lamps are prohibited.
	Area Plans: Forever North; Destination Midtown; Historic Florence; Joslyn Neighborhood; Benson-Ames, North Omaha Village Revitalization; South Omaha	NA	NA

	Development Project; Historic Boulevard System, 2013		
Houston city, TX	Code of Ordinances, Chapters 5 (Amusements), 28 (Miscellaneous Offenses and Provisions), 32 (Parks and Recreation), 40 (Streets and Sidewalks)	Glare, light trespass, distraction	5-102. Shading of Lights: Lights in outdoor amusement areas to be equipped with shades or hoods to prevent the same from shining directly into any residence 28-37. Attention-getting devices. Unlawful to place any attention-getting device on private or public property including device with blinking, rotating, moving, chasing, flashing, glaring, strobe, scintillating, search, flood or spot lights. • Display surfaces of signs not to contain any flashing lights. • Tower not to be artificially lighted unless required by law or regulation; design to minimize disturbance to any adjacent residence(s) or landmark 32-231. Park Vending. Lighting designed to prevent the source of light from being seen from any nearby property. Illumination not to be in excess of 0.5 f.c. beyond park boundaries. 40-10.1 Sidewalk cafes. shall not cause illumination to be obtrusive to adjacent property owners; illumination beyond the boundaries of the property not to exceed 0.5 f.c.
	Code of Ordinances, Chapter 9 (Aviation),	Glare, uplight	9-360. Land use restrictions. Prohibited: "uses that emit glaring light or employ highly reflective surfaces."
	Code of Ordinances, Chapters 26 (Parking), 33 (Planning & Development), 37 (Public Utilities), 38 (Railroads), 42 (Subdivisions, development and platting)	NA	NA
	General Plan, Plan Houston, adopted 2015	NA	NA
	Other relevant reports: • Acres Home Mobility Study Final Report • Full report – Benefits of the White Oak Bayou Greenway	NA	NA

	• Urban Houston Framework 2013 (pdf) • Feasibility Report: Expanding the Transit Corridor Ordinance Rules, 2009 (pdf) • Mercury Drive Study, 2009 • Lower Fifth Ward – Analysis for Underserved Communities, 2009 • Hillcroft-Harwin Circulation Study, 2009 • Washington Avenue Urban Studies, 2008 • Urban Corridor Planning, 2006-2009		
Seattle city, WA	Title 23-Land Use Code	Glare, light trespass, distraction	• External light to be shielded and directed away from adjoining properties, uses, rights-of-way, and nearby residential properties; screened with walls or landscape elements as applicable • When nonconforming exterior lighting is replaced, new lighting to conform to light and glare standards of the respective zone. • Director may limit intensity of illumination; change location. • In mixed residential-commercial districts, higher poles (40 ft) are permitted provided that "ratio of watts to area is at least 20% below the maximum exterior lighting level permitted by the Energy Code." • School athletic fields: higher poles (up to 100 ft) may be permitted where impacts from light and glare are minimized to the "greatest extent possible." Applicant must demonstrate reduction in impacts from light and glare, and that it has conducted a public workshop for nearby residents to solicit comments and suggestions. When the proposal extends the lighted area's duration of use, applicant must address and mitigate potential impacts.

			• Signs: Prohibited in all zones, with some exceptions (such as Arena subarea overlay zone): flashing signs, light festoons, searchlights, video display signs. • Arena subarea overlay zone. video displays to be limited to 500 nits (candela per sq. m.) between dusk and dawn; prohibited between 10 pm and 7 am (with some exceptions). • Shoreline modifications. Artificial night lighting shall first be avoided. If that is infeasible, lighting should minimize impacts on the aquatic environment to the maximum extent feasible. • Shoreline setback natural athletic fields allow no artificial lighting.
	Title 25: Environmental Protection and Historic Preservation	Light trespass	Minimize impacts by directing light away from wetlands.
	All other Titles in the City's Code of Ordinances	NA	NA
	Seattle 2035 Comprehensive Plan, adopted 2020 (and 2021 and 2022 amendments)	Glare, light trespass	LU 5.14. Establish controls on lighting and the glare from reflective materials to limit impacts on surrounding uses EL-P3. Minimize, through zoning regulations and/or design review potential for impacts on residential uses from artificial light or glare
	Climate Change and Vulnerability Assessment and Adaptation plan	NA	NA
	Seattle Light, Avian Protection Plan	NA[4]	NA
Los Angeles city, CA	Zoning Code	Glare, light trespass, distraction	• Lighting to be shielded so that the light source cannot be seen from adjacent residential properties • Ocean- Submerged land zone. Emission of artificial light should be controlled. • Sports fields: lighting standards may be built to maximum height of 75 ft. and must be equipped with appropriate beam control to direct light onto field and prevent glare and spill light from impacting neighboring properties, and shall include automatic timer with on/off scheduling system

[4] The program contains activities related to bird safety as it relates to nesting on poles, perching on wires, and tree and vegetation management.

		• Sign district, exterior site lighting. Maximum illumination of 0.2 horizontal and vertical f.c. at site boundary and 0.01 horizontal f.c 15 ft. beyond site. No more than 5% of initial designed lumens shall be emitted at an angle of 90 degrees or more from nadir. All fixtures shall be fully shielded. • Prohibited: Signs with flashing, mechanical and strobe lights. Maximum illumination of 3 f.c. above ambient lighting as measured at the nearest residential property line.
Zoning Code (Design Standards)	Darkness preservation	Scenic corridor: Lighting only at intersections and parking areas, and kept to a minimum useful intensity
Electrical Code		• 93.0107. Prohibited (with exceptions for public agency governed lighting): Distracting, confusing or nuisance lighting viewed from public streets used by vehicles. • 93.0117. Outdoor lighting affecting residential property. No more than 2 f.c. affecting abutting residential property, no direct glare; time limits for specific night time recreational uses affecting residential properties.
General Plan, Overview, 2017; Conservation Element, 2001; Open Space Element, 1973	NA	NA
Framework Element, 2001, Chapter 9—Infrastructure and Public Services	Glare, light trespass, light pollution	Street Lighting. "Unregulated, street lighting can contribute to negative factors in the community and the nighttime environment including glare, light trespass, and light pollution." Goal 9P(3). protect and preserve the nighttime environment, views, driver visibility, and otherwise minimize or prevent light pollution, light trespass, and glare. • 9.39.2 Ensure that lighting meets the minimum City standards and minimizes or prevents light pollution, light trespass, or glare • 9.40.2 Require parking lot lighting and related pedestrian lighting to meet recognized national standards • 9.40.3 Develop regulations to minimize or eliminate the adverse impact of lighting due to light pollution, light trespass, and glare for facade, security, and advertising lighting
Community Plans (Except those noted below)	NA	NA

	Community Plans for Brentwood Pacific Palisades, Central City North, and Encino Tarzana	Glare, light trespass	Standards related to shielding, directing light to target areas, directing away from adjacent residential properties
	Granada Hills-Knollwood Community Plan	Sky glow, light trespass	Goal CF15: A street-lighting system that protects and preserves the nighttime environment and contributes to appropriate levels of lighting, with minimum impact on the environment and adjoining properties.
	LA Amendment to the CA Green Building Code, 2020	Light pollution, glare, light trespass, darkness preservation	Interior and exterior lighting designed such that zero direct-beam illumination leaves building site. • Shield all exterior luminaires or provide cut off luminaires • No more than .01 f.c. to escape 15 ft beyond site boundary • Automatically control exterior lighting dusk to dawn to turn off or lower light during inactive periods
	LA Administrative Code, Street Lighting Improvements	NA	NA
	LA Lights Strategic Plan, 2020-2025	NA	NA

Table 4: Flagstaff Lighting Standards (updated 2022)

GENERAL OUTDOOR LIGHTING		**Lighting Zone 1**	**Lighting Zone 2**
Total light output permitted			
Commercial, industrial, multi-family		17,500 lumens/net acre	35,000 lumens/net acre
Single-family, duplex parcels		1,000 lumens/lot or parcel	5,000 lumens/lot or parcel
Shielding			
Commercial, industrial, multi-family	Class 1 and 2 lamps	Fully-shielded required	Fully-shielded required

	Class 3 lamps (less than 1,750 lumens per lamp)	Fully-shielded required	Partially shielded permitted (fully shielded preferred)
	Class 3 lamps (1,750 lumens or more per lamp)	Not permitted	Not permitted
Single-family, duplex parcels			
	Class 1, 2, 3 lamps(less than 700 lumens per lamp)	Fully-shielded required	Partially shielded permitted (fully shielded preferred)
	Class 1, 2, 3 lamps(700 lumens or more per lamp)	Not permitted	Not permitted
Fixture characteristics			
	Class 1	2700K or less CCT	2700K or less CCT
	Class 2	Narrow-spectrum Amber LED	Narrow-spectrum Amber LED
	Class 3	All light sources permitted	All light sources permitted
Curfew (Time limit)			
	Outdoor lighting, all classes (w Class 2 located more than 50 ft from structure)	No later than 30 mins after business closes, remain off until business reopens	No later than 30 mins after business closes, remain off until business reopens
	Outdoor lighting for sports facilities	9:00 PM	11:00 PM
SIGNS		**Lighting Zone 1**	**Lighting Zone 2**
Illumination		Treated as Class 1 lighting	
		only with steady, stationary, fully shielded light sources directed solely onto the sign without causing glare	

	Prohibited: signs with	Lighting devices with intermittent, flashing, rotating, … or any lighting effect creating the illusion of motion; laser or hologram lights; electronic displays; changeable copy LED signs.	Lighting devices with intermittent, flashing, rotating, … or any lighting effect creating the illusion of motion; laser or hologram lights; electronic displays; changeable copy LED signs.
Display characteristics		permitted: illuminated text and symbols permitted against non-illuminated background only	permitted: As noted for LZ1, or with generally lighter text and symbols against a colored (not white, off-white, light gray, cream, or yellow) background. Text and symbols may be white, off-white, light gray, cream, or yellow.
Curfew		9:00 PM (with timer switches required)	11:00 PM (with timer switches required)

Endnotes

[1] This is a unit of luminance, measured in candelas per square meter, in which the natural night sky measures at 1 mcd/m^2 (or 1 millicandela per sq. meter) and the night sky lit with the full moon at 1.4 mcd/m^2 (Source: en.wikipedia.org/wiki/Orders_of_magnitude_(luminance)). In comparison to lumens, which refer to the total amount of light emitted by a lighting apparatus, candela refers to the amount of light emitted by a lighting device in a particular direction. A light sphere that uniformly radiates one candela in all directions has a total luminous flux of 1 cd × 4π sr = 4π cd·sr or approximately 12.57 lumens (sr = steradian).

[2] Correlated-color (CCT) temperature is used to describe the *light appearance* provided by a light bulb. It is measured in degrees of Kelvin (K) on a scale from 1,000 to 10,000. Kelvin temperatures for commercial and residential lighting applications range from 2000K to 6500K, with the latter level of CCT typically associated with daylight. CCT is commonly used as a short hand to suggest the amount of blue light; however, this can be misleading. To produce the same amount of light, a 2300K source could emit more blue than one with 3000K (Schulte-Römer *et al* 2018, p. 59).

[3] For example, New York requires lamp CCT (a proxy for brightness but with implications for blue wavelength light and into the violet range) to be between 3000K and 4100K, with minimum color rendering index of 65. A color rendering index (CRI) is a quantitative measure of the ability of a light source to reveal the colors of various objects faithfully in comparison with an ideal or natural light source. Light sources with a high CRI are used in medical and artistic applications, where color is critical. Fluorescent lights range from about 50 to about 98. Typical LEDs have a CRI of 80 or more. (Source: https://en.wikipedia.org/wiki/Color_rendering_index).

[4] I am grateful to Dr. Christian Luginbuhl, astronomer, who patiently explained many aspects of outdoor lighting, developing objective codes to control for specific aspects, and his group's experience developing many iterations of the Lighting Code in Flagstaff.

[5] This term was found in codes with capitalization, i.e., NITS (Omaha), as well as without (Seattle). Interchangeable with *candelas per sq. meter*, a unit of luminance, or light emitted per unit area, it is frequently used to specify the brightness of a display device, such as a billboard with dynamic images. The Illinois Coalition for Responsible Outdoor Lighting (n.d.) recommends a maximum limit of 5,000 NITS during the daytime. It contends that a limit of 150 NITS surface brightness for digital signs is comparable to the nighttime signage that is widespread across the nation and is aligned with the recommendations of the Illuminating Engineering Society of North America. It accepts a higher limit of 200 NITS for a sign under certain conditions, a level that is lower than Omaha's (and Seattle's and Houston's) allowable maximum limit for such signs. Meanwhile, IDA's (2018) standards for illuminated signs for its Certified Dark Sky Community designation states, "Luminance levels between sunset and sunrise not to exceed 100 NITS (100 candelas per sq. meter) as measured under conditions of a full white display."

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